

Personality Prediction Based on Users' Tweets

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Abstract— The extensive use of social media has a tremendous impact on culture, business and politics in the world at large with potentially positive and negative effects. Personality traits are generally referred to as relatively stable patterns based on one's past experiences, upbringing, environment, all the opinions that one forms over time. It's closely bonded to how one feels, thinks and behaves, which has been associated with a wide range of important life outcomes and choices. Hence, in recent years, there has been a massive increase in the interest to develop models which use online data on human behaviour and preferences to automatically predict an individual's personality traits. In our research, to compute the accuracy of the personality predicted, it is first ensured that the dataset of stream of Tweets is almost equally divided into 16 different personality types, and be cleaned properly. We propose an advanced framework to identify and scrape tweets and retweets of Twitter user accounts and eventually applying Machine Learning algorithms for Personality Prediction.

Keywords—*MBTI model, Feature Extraction, TF-IDF vectorizer, XGBoost, Personality Prediction.*

I. INTRODUCTION

Twitter is one of the most used social media platforms not only by the citizens but the governments, world leaders, companies and organisations. Advances in consumer electronics (e.g., smartphones, wearables) and the subsequent development of mobile sensing methods have facilitated the collection of highly detailed, multi-dimensional data on behaviours and situations. Social media allows users to build an online persona through posting content such as text, images, links or through interaction with others. How users present themselves is a type of behaviour usually determined by differences in demographic or psychological traits. The behavioural residue harvested from websites and online social media platforms is also another valuable source of data on behaviour linked to personality traits. Twitter statistics [1] [2] [3] show that there are 199 million monetizable daily active users as of June 2021 and 500 million tweets sent per day.

With such a huge amount of data in the form of tweets and the whole banked history, we can incentivise it to predict a user's personality. The objective is to abstract key insights from the Twitter data stream and then align them with the key results. The growing number of social media users has resulted in a massive growth in the volume of online material. The information that these people share on

social media may frequently reveal a lot about their personalities.

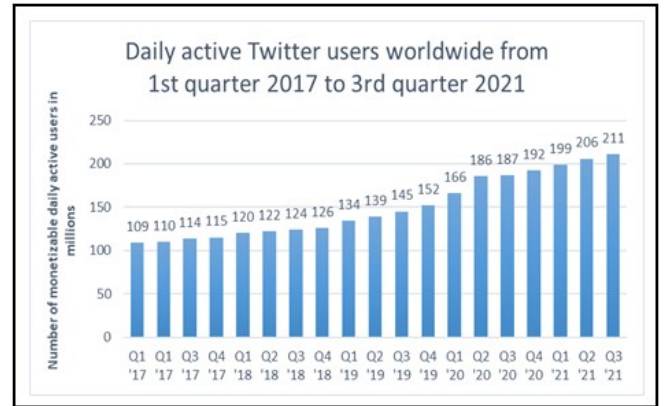


Fig. 1 Number of Twitter users worldwide in millions from 2017 to 2021.

II. RELATED WORK

There have been waves spread wide across the research communities, scientists, scholars, and even doctors to study the various personalities we humans possess. Not only that but also what have been the factors that have contributed to the particular personality of a human. There have been age-old questions that does a human's personality change over time with the various challenges and circumstances always hurled his way. [4], [5] have come a long way in determining the current personality based on user behaviour, reactions to certain situations, personal factors and influences, past experiences, likes and dislikes and many more. An abundance of research has been done in the field of personality prediction but has often been disregarded because of the data being biased, not enough sampled data which covers a large subset of the population involving all types of personality, models not proving to be accurate, privacy concerns when scraping the user's data from Social Media sites and many more.

In this paper [6], the authors have presented a personality prediction system for the region of Bahasa, Indonesia which is based on the Twitter User's information, in essence, the user's choice of words. They collected total of 359 data from various users based on certain criteria such as number of followers, following, tweets, retweets, etc. The authors of this paper have used two different classifiers for this study namely Support Vector Machine and XGBoost which are trained on the

same dataset and run on Weka and R respectively. Results showed that the accuracy of XGBoost (97.9962%) to be higher than that of Support Vector Classifier (76.2310%). The future scope of this paper lies in expanding the dataset which is achieved in the implementation of our research.

In this paper [7], the authors have done a thorough analysis and research on a person's psychology and the various social networks they're a part of. The paper and study give a brief idea on the correlation between a user's personality and their behaviour on social media platforms like Facebook. The authors have used the myPersonality dataset. Data Pre-Processing was done using OpenNLP and Features Extraction was done using LIWC & SPLICE. In Prediction, XGBoost outperformed the average baseline of all features. Using social network features for personality prediction, one can achieve a higher performance using linguistic features. The highest accuracy was achieved for extraversion. They were successful in predicting the personality with an average accuracy of 74.2% with XGBoost being the most accurate. Although, a larger sample dataset could be used to increase the accuracy of the prediction and hence overcoming the limitation small sample dataset used in their study.

In this work [8], the authors have used Facebook as the social media platform to gather users' data from their profiles to predict their personality (OCEAN model). They have used a Selenium web driver to scrape the statuses and then this data was converted to vectors using TF-IDF vectorizer to remove unwanted data. For Classification and Prediction, they have used Ensembling Technique, and for Training and Prediction, Random Forest Classifier and Random Forest Regressor are used because of their binary, categorical, binary and numerical features and its significantly faster prediction speed. These models are trained on the myPersonality dataset which contains 5000 positive and 5000 negative statuses corresponding to personality traits in the OCEAN model. The mean squared error of the Random Forest Regressor was found to be 5.25 and the accuracy of the Random Forest Classifier was 64.24%. The future scope of this paper lies in removing the language barriers and identifying the fake user's profile and fake statuses of the user.

The authors in [9] opted to create their own dataset using the Twitter Streaming API. Their whole prediction model mainly focuses on the famous Big Five Model. The data was specifically pre-processed to avoid further complications during the implementation phase. The pre-processed data was later visualized using Python libraries. Eventually Classification algorithms like AdaBoost, Multinomial Naive Bayes and LDA were used. They came to a conclusion that Multinomial Naive Bayes had the highest accuracy of 73.43, precision of 0.7, recall of 0.71 and F1 score of 0.72. They similarly aim to use a multimodal approach in their future research which is something which hasn't been worked on yet.

The authors in [10] have discovered that Twitter users who are popular and listeners are strongly correlated with Neuroticism and Extraversion personality traits and the users who are saved by other users in their read-list are highly correlated with Openness trait, whereas influential users show the high tendency of Extraversion and Conscientiousness traits. [11] used ZeroR and Gaussian Processes Machine Learning algorithms to predict personality (OCEAN Model). In [12], The Dark Triad Personality traits (Psychopathy, Machiavellianism, Narcissism) are predicted using Support Vector Machine, Random Forest, J48, Naive Bayes Classifier algorithms on WEKA.

III. THE FOUR FACTOR MBTI MODEL

The human personality is a diverse and colourful phenomenon, ranging from quirky and reclusive to passionate and assertive. A person's particular patterns of thinking, feeling, and behaving are referred to as personality. It is the result of a combination of natural tendencies and inclinations, as well as external circumstances and experiences. Although one's personality might vary during one's life, one's key personality traits tend to remain stable into maturity.

The Myers-Briggs Personality Type Indicator (MBTI) is a technique that assists people in understanding their personality type, their strengths and weaknesses, their communication preference and how they communicate with others. Knowing the MBTI personality type helps adjust one's interpersonal approach to various situations and audiences.

In [13], it is mentioned that Carl Jung thought that a person's personality is made up of their preferences, or how they chose to do things. He hypothesised that four pairs of contrasting behaviour depict one's openness (Extroversion vs. Introversion), perception to information (Sensing vs. Intuition), decisions (Thinking vs. Feeling) way of living life (Judging v. Perceiving).

Preferences from each of the four pairs make up the Personality type. There are 16 types of Personalities in all; they are ISTJ, ISTP, ISFJ, ISFP, INTJ, INTP, INFJ, INFP, ESTJ, ESTP, ESFJ, ESFP, ENTJ, ENTP, ENFJ, ENFP.

As mentioned in [14], The INFJ personality type is found to be scarce among individuals carrying a weightage of only 1.5%. The INFJ group of individuals who possess the Introverted, Intuitive, Feeling and Judging type of a personality are the rarest on the planet. These folks are often perceived to be quiet, reserved and someone who hold back their feelings, don't often express it and are also emotionally less intelligent towards others.

According to the 16 Personalities Institute, ISFJ, the Defender personality type is the most common. This group accounts for 13.8 per cent of the population in the United States. This type of personality is quite common and there is a high chance that they are probably someone in your house, close friends, colleagues or someone you interact with quite often. They are most often referred to

as social introverts who are like most introverts but also have a sense of being involved with the community. They don't like to take any credits for their exemplary work and pretty much work behind the scenes thinking about the welfare of others and their organization. The people with these traits, are more often found in females and not men.

IV. METHODOLOGY

A. Analysis of the Dataset:

The MBTI dataset used in this study is obtained from Kaggle. This dataset contains 8675 streams of tweets and the corresponding MBTI Personality type of different Twitter users. Figure 2 shows that 21.1% of the users in dataset possess INFP Personality type and 0.447% of the users in the same dataset have the Personality type of ESTJ.

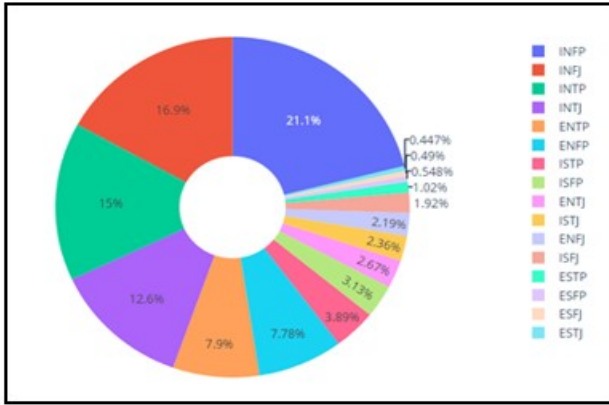


Fig. 2 Distribution of MBTI Personality types

B. Data Pre-processing

• Data Cleaning:

In this paper, we are working with a stream of tweets from several users, which is text data. They contain a variety of redundant data, such as hashtags and links, which are not helpful in prediction the personality and this may result in erroneous findings. As a result, we started off by removing all of the hashtags and links from the data. Then we deleted all the unnecessary symbols such as '?', '!', etc., and various emoticons. As a result, our data was cleaned and prepared for further processing.

• Data Splitting:

As illustrated in Figure 2, the dataset contains a large number of samples for some personality types and a small number of samples for others. As a result, training a classification model on such data would be inefficient. It would give biased results in favor of INFP, INFJ, and INTP, which collectively accounts for 53% of the dataset. A model trained on such dataset would hardly predict any personality type whose contribution to it is lesser than 2%. MBTI personality type is determined by four separate personality attribute comparisons. Each

comparison contains two contrasting personality traits.

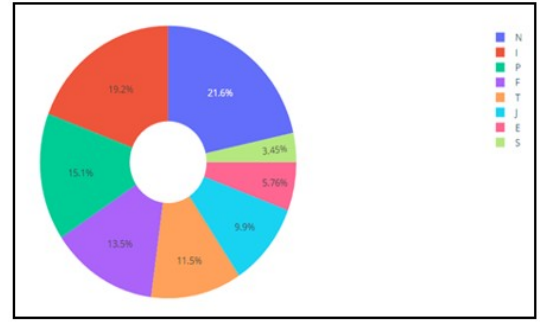


Fig. 3 Proportion of each personality type in the dataset

As a result, we divided each personality type into four separate personality characteristics, such as INFP has been subdivided into I, N, F, and P. Figure 3 depicts the distribution of personality traits in the resulting dataset, which has 34,700 rows.

C. Feature Extraction

To apply morphological analysis to words, lemmatization is used. It extends beyond word reduction and evaluates a language's entire lexicon. It is often viewed as significantly more informative than basic stemming. It uses the context of the surrounding text to identify the part of speech of a particular word. Feature extraction here is done using Term Frequency – Inverse Document Frequency (to be referred to as TF-IDF hereinafter) technique which is one of the most essential approaches for representing how relevant a certain word or phrase is to a given text in terms of retrieving the information. Term Frequency (TF) shows how important the word is to the given text/dataset/document with the assumption that the higher the frequency of the word, the higher will be its importance. Inverse Document Frequency (IDF) shows how much the word is relevant actually. It's not always that a phrase or word that appears often in the text/dataset/document such as stop-words be relevant. Since the stop-words obscure the context, they should be removed. The measure TF-IDF thus supplies us numeric values for the entire document.

D. Training

The initial unsampled data was pointing us in a direction of biasedness by predicting the personalities which had a larger chunk of values in the dataset. As a consequence, the personalities which had only a few values in the dataset were not being predicted. Thus, to avoid biased classification, the dataset was first sampled and then divided into training and testing sets for each of the four classifications. For each classification, four distinct and powerful classifiers were used namely: Logistic Regression, Multinomial Naive Bayes, Support Vector Classifier and XGBoost Classifier. The trained models were eventually evaluated using the testing data.

V. RESULTS

Our approach was tested using a holdout set of 20% of the dataset size and 10-fold cross-validation.

TABLE I. EVALUATION REPORT FOR I VS E

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	83.65	76.75	80.88	80.05
Multinomial Naive Bayes	76.13	75.75	76	75.94
XGBoost	83.11	77.5	80.88	80.2
Support Vector Classifier	87.25	77	82.88	81.80

TABLE II. EVALUATION REPORT FOR S VS N

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	79.26	81.25	80	80.24
Multinomial Naive Bayes	78.42	78.75	78.54	78.58
XGBoost	80.24	82.91	81.25	81.55
Support Vector Classifier	80.49	80.83	80.62	80.66

TABLE III. EVALUATION REPORT FOR T VS F

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	86.17	86.26	85.07	86.21
Multinomial Naive Bayes	78.03	89.66	80.75	83.44
XGBoost	83.10	83.28	81.79	83.19
Support Vector Classifier	86.12	85.94	84.9	86.03

TABLE IV. EVALUATION REPORT FOR J VS P

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	78.72	75.40	77.53	77.02
Multinomial Naive Bayes	68.64	70.74	69.24	69.67
XGBoost	81.94	75.98	79.64	78.85
Support Vector Classifier	79.96	74.38	77.89	77.07

TABLE V. 10-FOLD CROSS-VALIDATION REPORT FOR I VS E

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	80.51	80.12	80.21	80.16
Multinomial Naive Bayes	75.85	75.71	75.71	75.67
XGBoost	78.3	78.21	78.21	78.2
Support Vector Classifier	80.11	79.71	79.71	79.64

TABLE VI. 10-FOLD CROSS-VALIDATION REPORT FOR S VS N

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	83.3	83.15	83.15	83.13
Multinomial Naive Bayes	80.11	79.92	79.92	79.88
XGBoost	79.21	79.13	79.13	79.12
Support Vector Classifier	83.65	83.52	83.52	83.5

TABLE VII. 10-FOLD CROSS-VALIDATION REPORT FOR T vs F

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	86.1	86.1	86.09	86.08
Multinomial Naive Bayes	80.26	79.52	79.52	79.9
XGBoost	83	82.99	82.99	82.98
Support Vector Classifier	85.89	85.88	85.88	85.87

TABLE VIII. 10-FOLD CROSS-VALIDATION REPORT FOR J vs P

Algorithm	Precision %	Recall %	Accuracy %	F1 Score %
Logistic Regression	79.75	79.65	79.65	79.63
Multinomial Naive Bayes	72.53	72.5	72.5	72.5
XGBoost	79.32	79.12	79.12	79.08
Support Vector Classifier	78.51	78.45	78.45	78.44

Tables I, V exhibit the evaluation and cross-validation report for classification of Introversion and Extroversion personality traits using four different classifiers; Logistic Regression, Multinomial Naive Bayes, XGBoost, and Support Vector Classifier. The Support Vector Classifier (SVC) has the highest accuracy of 82.88% in the evaluation report. Whereas Logistic Regression generates greater results in a 10-fold cross-validation report, with a cross-validation accuracy of 80.21%.

Support Vector Classifier gives the highest ratio between true positives and all positives, i.e. precision score of 80.49%, for classification between Sensing (S) and Intuition (N), whereas XGBoost classifier fared best with 82.91% recall and accuracy of 81.25%, as shown in Table II. Multinomial Naive Bayes had the lowest accuracy of 78.54%. However, in the 10-fold cross-validation report provided in Table VI, the XGBoost classifier had the lowest scores with a cross-validation accuracy of 79.13% and the SVC had the highest scores with a cross-validation accuracy of 83.52%.

Table III displays the evaluation report for the classification of Thinking(T) and Feeling(F) personality traits. Logistic Regression fared the best with an accuracy score of 85.07%, followed by SVC with an accuracy score of 84.9%, XGBoost classifier with an accuracy score of 81.79%, and Multinomial Naive Bayes classifier with an accuracy score of 80.75%. However, the Multinomial Naive Bayes classifier achieved the highest recall (89.66%). In the 10-fold cross-validation result presented in Table VII, Logistic Regression once again surpassed all other classification algorithms with a cross-validation accuracy of 86.09%.

Table IV displays the evaluation report for the classification of Judging(J) and Perceiving(P) personality traits. XGBoost proves to be the most accurate with an accuracy of 86.09%. It also outperforms the other classification algorithms in precision, recall, and F1 score. In the 10 fold cross-validation result presented in Table VIII, Logistic Regression has the highest accuracy of 79.65%. It also outperforms the other classification algorithms in other factors like precision, recall and F1 score.

From 10-Fold Cross Validation Report as mentioned in above tables, we can deduce that, for all the factors, Logistic Regression algorithm performs best in the prediction of I vs E, T vs F and J vs P personalities, whereas Support Vector Classifier performs best for the prediction of S vs N personality trait.

We infer from the Evaluation Report that XGBoost algorithm performs best for all the factors in J vs P personality prediction. Whereas Support Vector Classifier, XGBoost and Logistic Regression holds good in the prediction of I vs E, S vs N and T vs F personality traits respectively.

VI. CONCLUSION

A generalized framework can be created to predict the user's behaviour based on the tweets. The study finds that testing efficient Machine Learning algorithms and eventually picking out the most accurate one can help in determining the precise personality type of the user. Personality prediction on Twitter presents unique characteristics and challenges that make existing detection algorithms not entirely accurate along with the unbiased datasets. Because of the large amounts of data produced in the form of tweets, posts, videos which lead to trends, news, fake news, information, retweets, likes, comments, it has become really important to retrieve this data ethically without bias and eventually gain insights about the person's personality. In this survey, we present a comprehensive review of predicting a user's behaviour based on his tweets, existing algorithms from a data mining perspective, evaluation metrics and representative datasets.

From the 10-Fold Cross-Validation Report as mentioned in the above tables, we can deduce that, for all the factors, the Logistic Regression algorithm performs best in the prediction of I vs E, T vs F and J vs P personalities with the accuracy of 80.21%, 86.09%, 79.65%, whereas the Support Vector Classifier performs

best for the prediction of S vs N personality trait with the accuracy of 83.15%.

We infer from the Evaluation Report that the XGBoost algorithm performs best for all the factors in J vs P personality prediction. Whereas Support Vector Classifier, XGBoost and Logistic Regression holds good in the prediction of I vs E, S vs N and T vs F personality traits respectively.

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